Improve Generalization

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Improve Generalization before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Feature Engineering

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Overfitting

Without opening the old lesson, explain one key idea from **Overfitting**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Improve Generalization trace and record the first step you can explain without help.

Use W2: Interpret a gap for **Overfitting** and model **capacity**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Improve Generalization explanation against the supplied facts before discussing it.

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Spotted leaves have blue backgrounds and healthy leaves white backgrounds in **training**. Design a generalisation investigation with fresh evaluation.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Improve Generalization target to a trace and a justified conclusion.

All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Improve Generalization case and show the deciding evidence.

A classifier relies on tray colour; new trays reduce performance.

Propose controlled improvement.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A team proposes more representative **training** examples and a simpler model. How should it compare changes while protecting the final test?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 30 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Improve Generalization adds to Responsible Plant Classifier and keep the evidence.

Improve Plant Classifier with one approved change and a full before/after **evidence** table.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Select from evidence

> Select from evidence. Apply the supplied Improve Generalization facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Improve Generalization", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Choose among the three worked candidates and explain the **training** trap.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Interpret a gap

> Interpret a gap. Apply the supplied Improve Generalization facts and keep your reasoning.

Does a **training**-validation gap alone prove **overfitting**?

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Propose a controlled change

Propose a controlled change. Apply the supplied Improve Generalization facts and keep your reasoning.

Describe one fair next development comparison.

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Improve Generalization facts and keep your reasoning.

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

W5 Limit a diagnosis

> Limit a diagnosis. Apply the supplied Improve Generalization facts and keep your reasoning.

What extra **evidence** would you inspect before concluding a model memorized noise?

My reasoning and evidence

WORKBOOK · 17 OF 26

Distribution shift — W1 Compare conditions

> Use the supplied Improve Generalization task to record your reasoning, evidence and next step.

Calculate each worked error rate and the difference.

My reasoning and evidence

WORKBOOK · 18 OF 26

Distribution shift — W2 Read pooled evidence

> Use the supplied Improve Generalization task to record your reasoning, evidence and next step.

Why is 8/40 incomplete by itself?

My reasoning and evidence

WORKBOOK · 19 OF 26

Distribution shift — W3 Wait for labels

Use the supplied Improve Generalization task to record your reasoning, evidence and next step.

Can changed **prediction**s alone prove an accuracy drop?

My reasoning and evidence

WORKBOOK · 20 OF 26

Distribution shift — W4 Apply independently

Use the supplied Improve Generalization task to record your reasoning, evidence and next step.

Old-like data has 3 errors/30 and new-condition data 5/10. Give both rates and the pooled rate.

My reasoning and evidence

WORKBOOK · 21 OF 26

Distribution shift — W5 Plan a response

Use the supplied Improve Generalization task to record your reasoning, evidence and next step.

State a safe classroom response to **input**s outside the supported range.

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

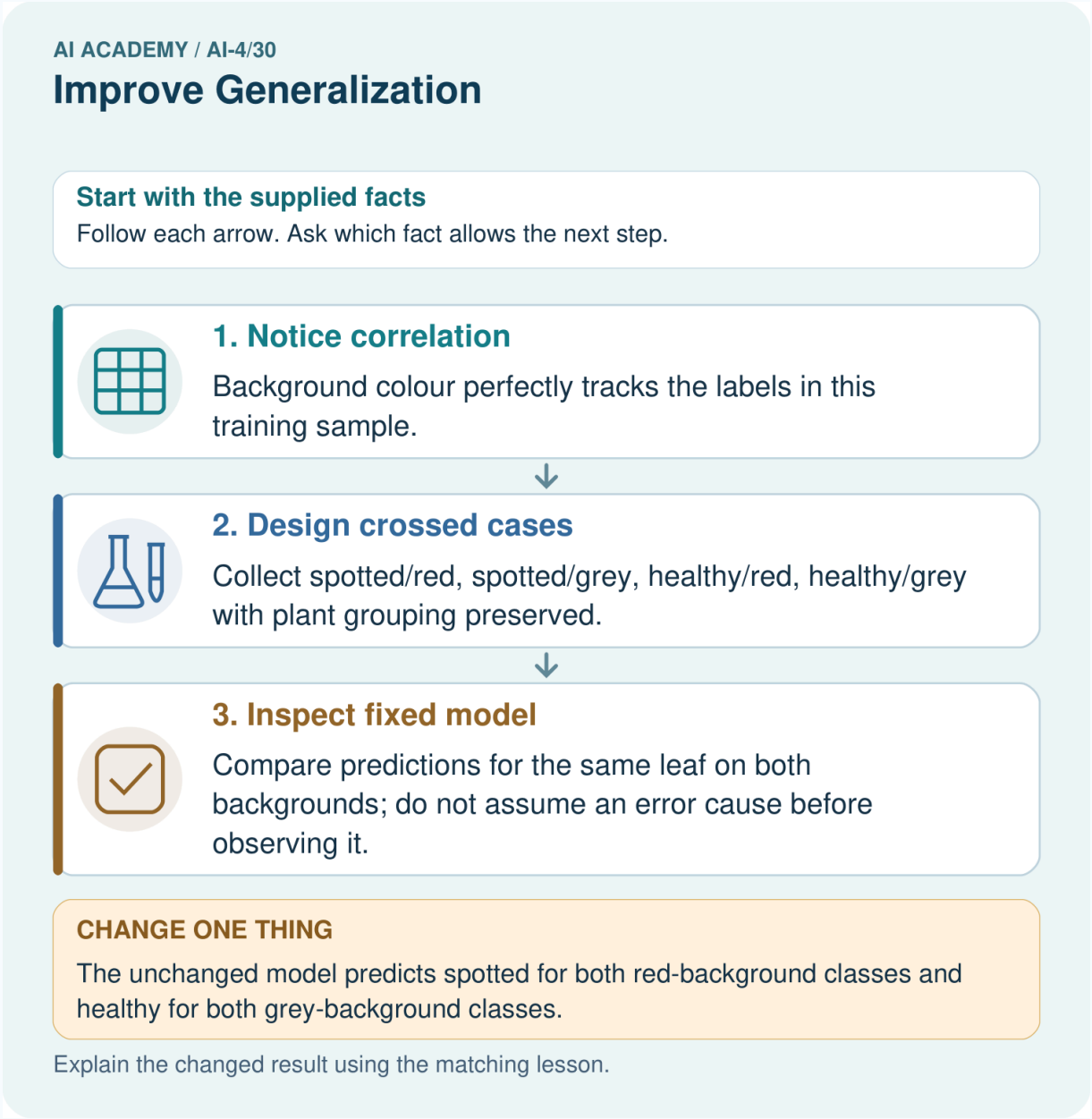
Pictorial case practice: Improve Generalization

> Test whether a plant model follows the leaf or a convenient background shortcut. Explain the labelled step before checking the changed condition.

Test whether a plant model follows the leaf or a convenient background shortcut.

Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards. Intended use allows both backgrounds for either class.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Test whether a plant model follows the leaf or a convenient background shortcut.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

1. Notice correlation

2. Design crossed cases

3. Inspect fixed model

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect fixed model Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. Read the picture in order.

Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Which conclusion is supported by the supplied facts?

1. It fails on spotted/grey and healthy/red in the crossed audit.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: It fails on spotted/grey and healthy/red in the crossed audit. Those supplied **prediction**s follow background rather than the intended leaf target.

Repair a missing building block

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Overfitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model **capacity**. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

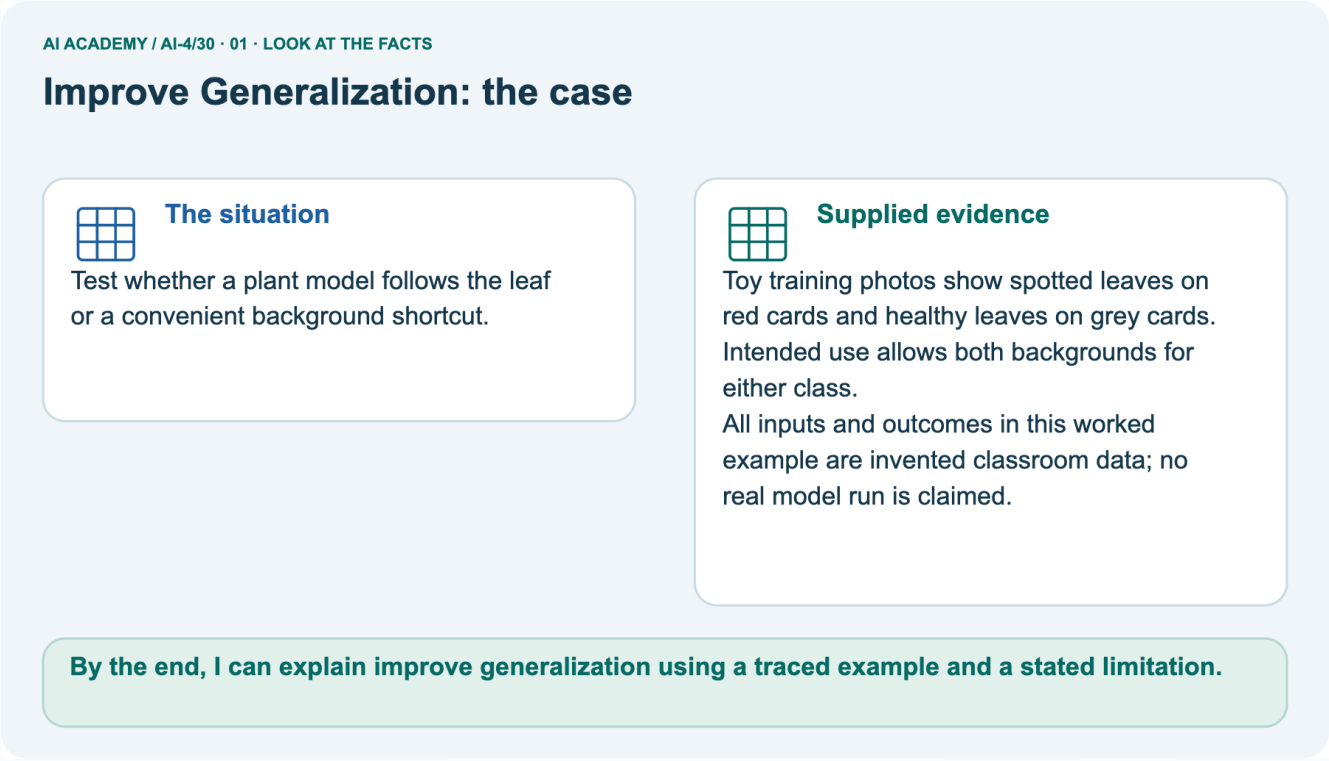
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Improve Generalization picture from its labels, then record a relevant limit.

01. Improve Generalization: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

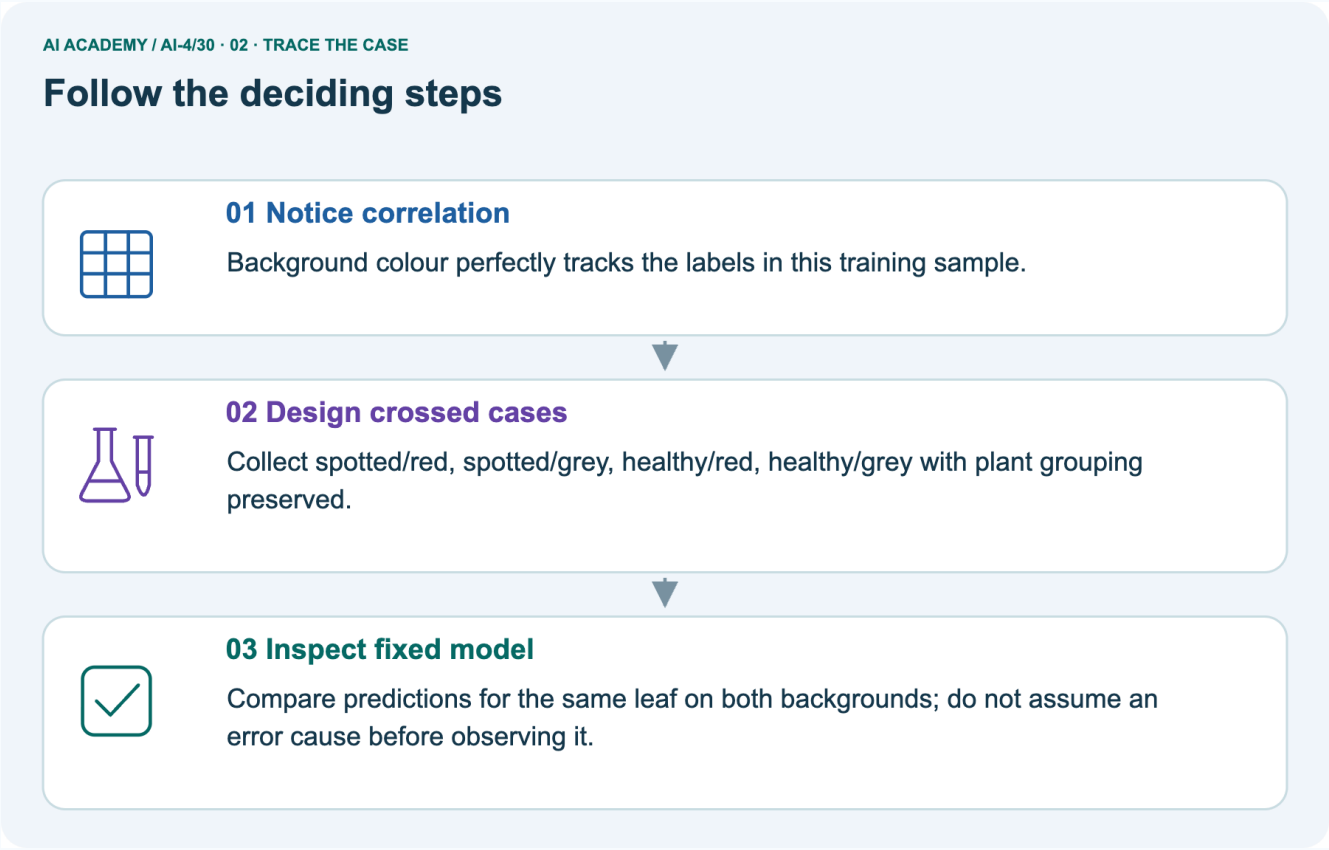
Test whether a plant model follows the leaf or a convenient background shortcut. Supplied facts: Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards.

Intended use allows both backgrounds for either class. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



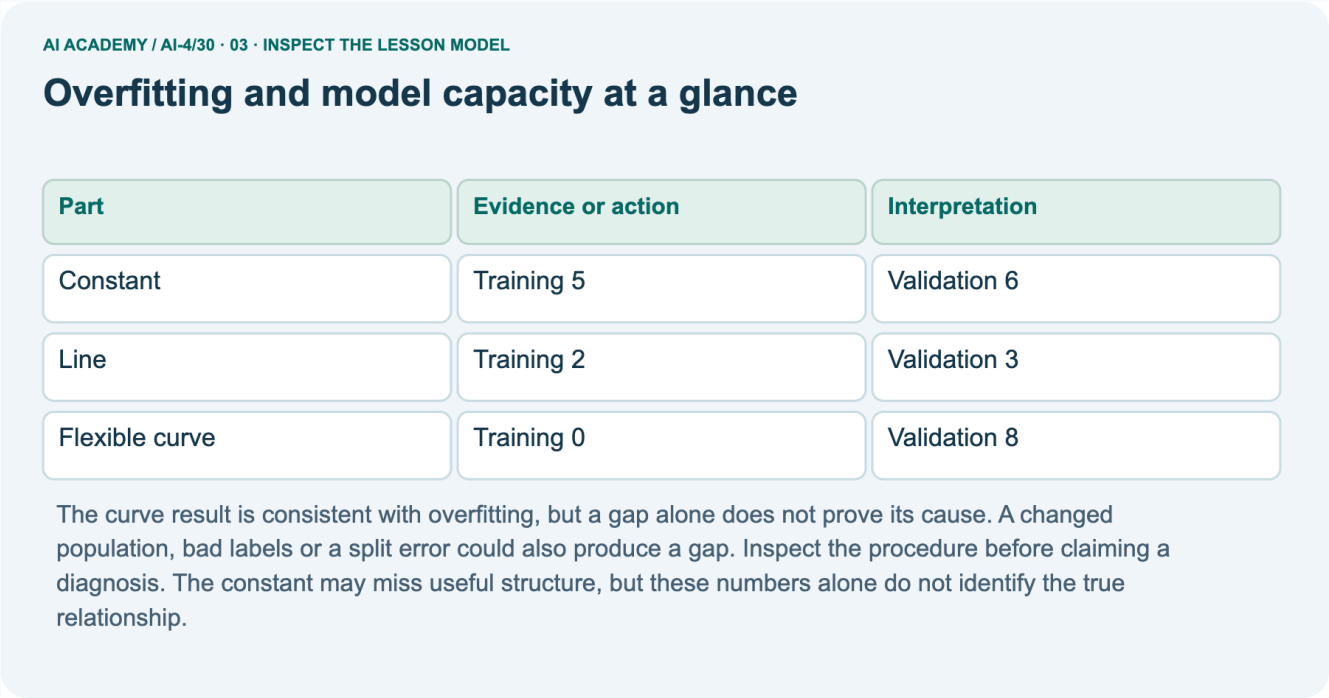
Follow the labelled picture with the matching explanation. Open the vector diagram

Notice correlation: Background colour perfectly tracks the labels in this **training** sample. → Design crossed cases: Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved. → Inspect fixed model: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Overfitting and model capacity at a glance



AI ACADEMY / AI-4/30 · 03 · INSPECT THE LESSON MODEL

Overfitting and model capacity at a glance

Part	Evidence or action	Interpretation
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

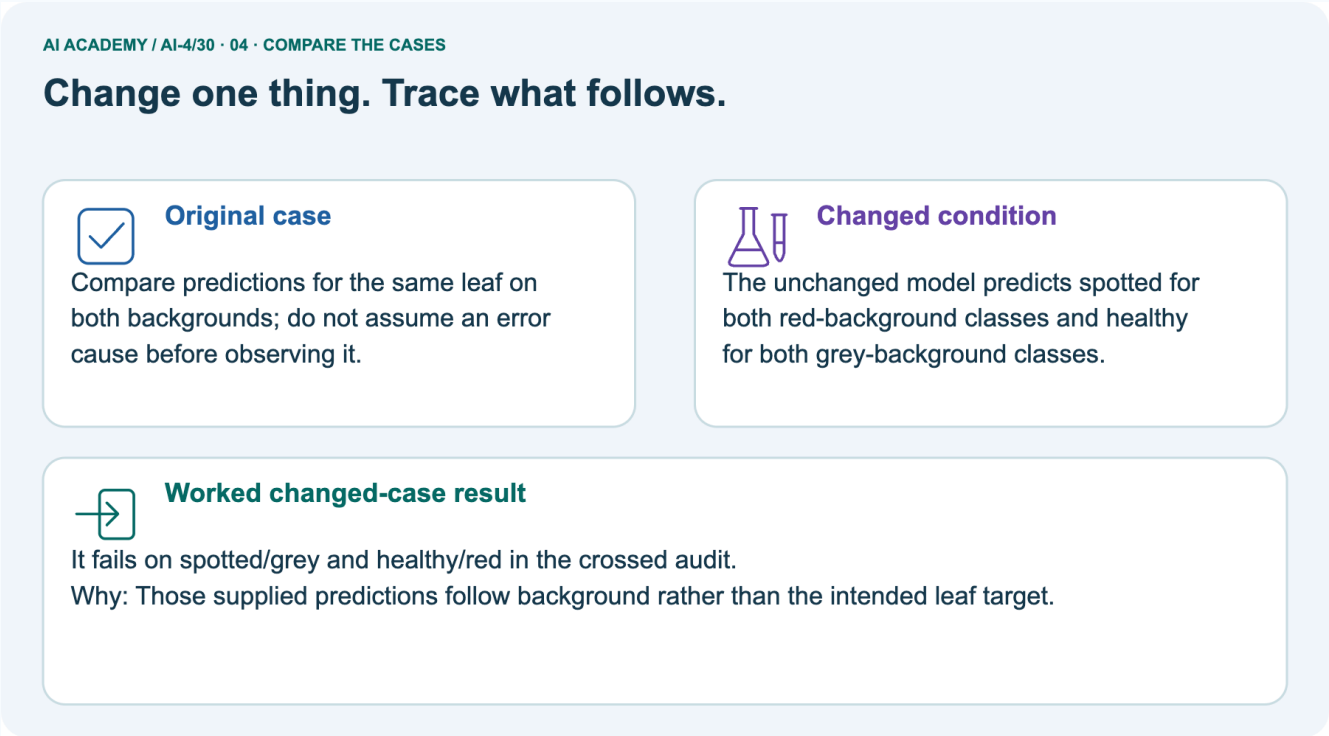
Follow the labelled picture with the matching explanation. Open the vector diagram

Overfitting and model **capacity** at a glance. Constant | **Training** 5 | Validation 6. Line | **Training** 2 | Validation 3. Flexible curve | **Training** 0 | Validation 8

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

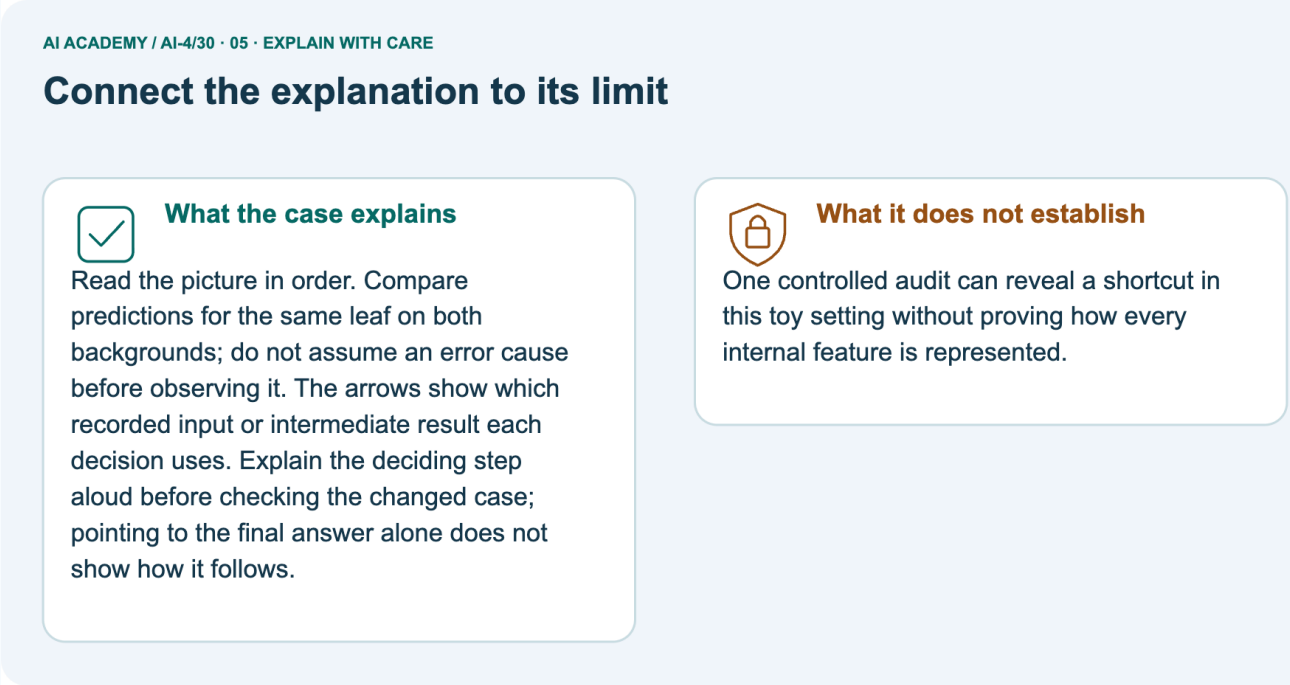
Changed condition: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Expected result: It fails on spotted/grey and healthy/red in the crossed audit.

Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

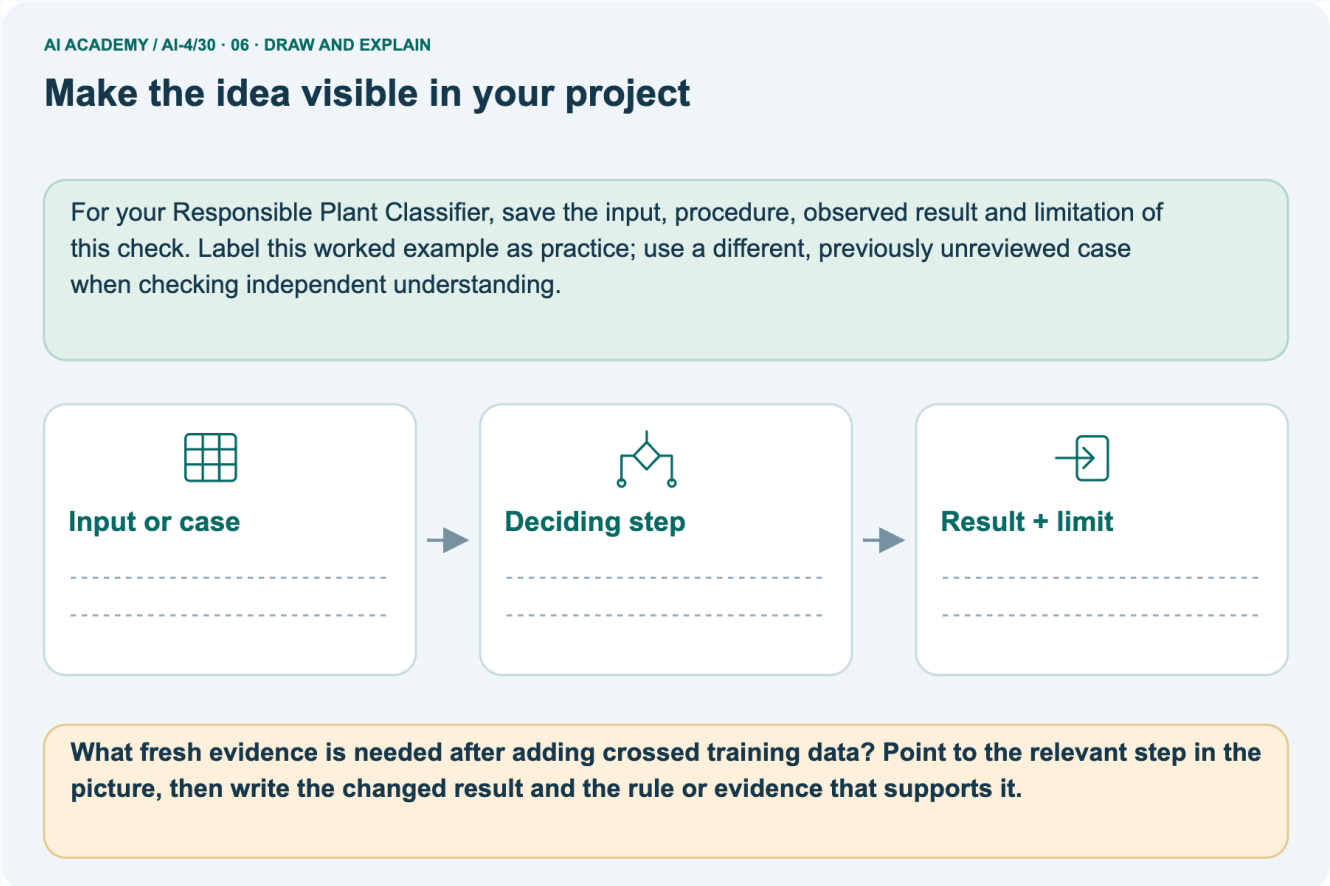
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.